DRAFT INSPECTION AND TEST PLAN — GENERATED BY VR INSPECTOR

Concrete-encased duct bank — 4-way 4 in PVC

Inspections and tests for an underground concrete-encased duct bank, from trench to backfill and records.

16 lines · 1 hold point · generated Nov 13, 2026 09:00 UTC · draft D3A0C0DE

Draft — generated by VR Inspector; review before use.
This is an AI-generated starting point, not an engineered document. It makes no claim of engineering adequacy or code compliance. A competent person must check every line, criterion and reference against the project's drawings, specifications and governing codes before it is issued or used.

SCOPE OF WORK AS DESCRIBED

> Install a 4-way 4 in PVC duct bank, concrete encased, 300 ft from the substation to the MCC building, with pull boxes every 100 ft, then backfill and compact.

ASSUMPTIONS AND THINGS TO CHECK

- Assumed one straight run with two pull boxes; add lines for each pull box if the drawings show more.
- Burial depth, concrete cover, slump and compaction depend on the project — fill them in from the drawings and specifications.
- Confirm every value against the project documents before issue.

SIGN-OFF ROLES

Contractor | Owner's representative | Third-party inspector

Type: **H** Hold point — work stops until released · **W** Witness point — notify, may proceed if not attended · **R** Review of documents · **I** Information / surveillance

Before work

NO.	TYPE	CHECK AND ACCEPTANCE CRITERION	SIGNED BY · RESULT
1	R	**Conduit, spacer and fitting submittals accepted** **Criterion:** Accepted submittals on file for conduit, spacers, couplings, bends and pull boxes before material is installed **Reference:** Project specifications **Record:** Submittal log	Owner's representative Pass / Fail / N/A _____
2	R	**Concrete mix design accepted** **Criterion:** Accepted mix design on file; strength and color dye as specified **Reference:** ACI 301 **Record:** Mix design submittal	Owner's representative Pass / Fail / N/A _____
3	W	**Route staked and existing utilities located** **Criterion:** Alignment and grades staked; all crossings marked and potholed where shown **Reference:** Approved drawings **Record:** Survey record, locate tickets	Contractor, Owner's representative Pass / Fail / N/A _____

Trench and conduit installation

NO.	TYPE	CHECK AND ACCEPTANCE CRITERION	SIGNED BY · RESULT
4	W	**Trench depth below finished grade** **Criterion:** **Project-specific value required — verify before issue** **Reference:** NFPA 70 **Record:** Depth readings every 50 ft **Record a number (in)**	Third-party inspector Pass / Fail / N/A _____
5	I	**Trench bottom firm and free of rock and water** **Criterion:** No standing water, no rock or debris larger than 1 in on the bed **Reference:** Project specifications **Record:** Daily report	Contractor Pass / Fail / N/A _____
6	W	**Conduit size, count and spacer spacing** **Criterion:** Four 4 in conduits per drawing; spacers at the interval shown, conduits held in position **Reference:** Approved drawings **Record:** Installation checklist	Third-party inspector Pass / Fail / N/A _____
7	I	**Joints glued and bends within allowed total per run** **Criterion:** Every joint solvent-welded; total bends between pull points within the allowed total **Reference:** NFPA 70 **Record:** Installation checklist	Contractor Pass / Fail / N/A _____
8	I	**Conduits anchored against flotation** **Criterion:** Tie-downs at the interval shown; no conduit lifted before or during the pour **Reference:** Approved drawings **Record:** Photo **Photo required**	Contractor Pass / Fail / N/A _____

Before encasement

NO.	TYPE	CHECK AND ACCEPTANCE CRITERION	SIGNED BY · RESULT
9	H	**Duct bank released for concrete encasement** **Criterion:** Lines 4–8 complete and accepted; conduit ends capped; mandrel test passed on every conduit **Reference:** Project specifications **Record:** Pre-pour release, photos **Photo required**	Owner's representative, Third-party inspector Pass / Fail / N/A _____
10	W	**Concrete cover around outer conduits** **Criterion:** **Project-specific value required — verify before issue** **Reference:** Approved drawings **Record:** Measured cover **Record a number (in)**	Third-party inspector Pass / Fail / N/A _____

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Encasement and backfill

NO.	TYPE	CHECK AND ACCEPTANCE CRITERION	SIGNED BY · RESULT
11	R	**Concrete delivery tickets checked** **Criterion:** Mix matches the accepted design; water added recorded; time since batching within the specified limit **Reference:** ASTM C94 **Record:** Batch tickets	Third-party inspector Pass / Fail / N/A _____
12	W	**Slump** **Criterion: Project-specific value required — verify before issue** **Reference:** ASTM C143 **Record:** Field test report **Record a number (in)**	Third-party inspector Pass / Fail / N/A _____
13	I	**Warning tape placed above the duct bank** **Criterion:** Continuous tape at the depth shown, full width of the bank **Reference:** Approved drawings **Record:** Photo **Photo required**	Contractor Pass / Fail / N/A _____
14	W	**Backfill compaction** **Criterion: Project-specific value required — verify before issue** **Reference:** ASTM D1557 **Record:** Density test report **Record a number (%)**	Third-party inspector Pass / Fail / N/A _____

Completion and records

NO.	TYPE	CHECK AND ACCEPTANCE CRITERION	SIGNED BY · RESULT
15	W	**Mandrel pulled through every conduit after backfill** **Criterion:** Mandrel passes end to end in every conduit; pull string left in place **Reference:** Project specifications **Record:** Mandrel test record	Owner's representative Pass / Fail / N/A _____
16	R	**As-built route and depths recorded** **Criterion:** As-built drawing shows alignment, depths and crossings **Reference:** Project specifications **Record:** As-built drawing	Owner's representative Pass / Fail / N/A _____

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